Book 1 — Lecture 2 — Exercise 2

Second Derivatives of Four Introductory Functions

Prompt. The derivative of a derivative is called the second derivative and is written $(d^2 f(t))/d^2$.[1] Take the second derivative of each of the functions listed above:

$$\begin{aligned} f(t) &= t^4 + 3t^3 - 12t^2 + t - 6, \\ g(x) &= \sin(x) - \cos(x), \\ \theta(\alpha) &= e^\alpha + \alpha \ln(\alpha), \\ x(t) &= \sin^2(t) - \cos(t). \end{aligned}$$

$f(t)$

From Exercise 1, $f'(t) = 4t^3 + 9t^2 - 24t + 1$.[2] Differentiate again:[3]

$$f''(t) = \frac{d^2 f}{dt^2} = 12t^2 + 18t - 24.$$

$g(x)$

We had $g'(x) = \cos(x) + \sin(x)$.[4] Thus

$$g''(x) = \frac{d^2 g}{dx^2} = -\sin(x) + \cos(x).$$

[5]

$\theta(\alpha)$ $(\alpha > 0)$

We had $\theta'(\alpha) = e^\alpha + \ln(\alpha) + 1$. Differentiating once more term by term,[6]

$$\theta''(\alpha) = \frac{d^2\theta}{d\alpha^2} = e^\alpha + \frac{1}{\alpha}.$$

[1] Operationally, the second derivative is "the derivative of the derivative": one first computes $f'(t)$, then differentiates that new function of t again. Standard Leibniz notation writes the variable in the denominator twice, e.g. $\frac{d^2 f}{dt^2}$ for a function of t, and $\frac{d^2 g}{dx^2}$, $\frac{d^2\theta}{d\alpha^2}$, $\frac{d^2 x}{dt^2}$ for the other independent variables below. The d^2 in the numerator indicates that differentiation has been applied twice.

[2] Exercise 1 computed first derivatives of the same four functions.

[3] For a polynomial, each differentiation lowers the highest power by one; here f was degree 4 in t, so f' is degree 3 and f'' is degree 2.

[4] As in Exercise 1, angles are in radians.

[5] Apply $(\cos x)' = -\sin x$ and $(\sin x)' = \cos x$ to $g'(x)$.

[6] For $\alpha > 0$, $\frac{d}{d\alpha}\ln(\alpha) = 1/\alpha$. The summand $+1$ is constant and has derivative 0. Also $(e^\alpha)' = e^\alpha$.

$x(t)$

One convenient form of the first derivative is $x'(t) = \sin(t)\big(2\cos(t) + 1\big)$. Differentiate using the product rule:[7]

$$\begin{aligned} x''(t) = \frac{d^2x}{dt^2} &= \cos(t)\big(2\cos(t) + 1\big) + \sin(t)\big(-2\sin(t)\big) \\ &= 2\cos^2(t) + \cos(t) - 2\sin^2(t) \\ &= 2\cos(2t) + \cos(t), \end{aligned}$$

using $\cos^2 t - \sin^2 t = \cos(2t)$.[8]

Result. In summary,

$$\begin{aligned} \frac{d^2 f}{dt^2} &= 12t^2 + 18t - 24, \\ \frac{d^2 g}{dx^2} &= \cos(x) - \sin(x), \\ \frac{d^2\theta}{d\alpha^2} &= e^\alpha + \frac{1}{\alpha} \quad (\alpha > 0), \\ \frac{d^2 x}{dt^2} &= 2\cos(2t) + \cos(t). \end{aligned}$$

Example. At $t = 0$: $f''(0) = -24$, $x''(0) = 2 + 1 = 3$. At $\alpha = 1$: $\theta''(1) = e + 1$.

[7] Write $x'(t) = u(t)v(t)$ with $u = \sin t$ and $v = 2\cos t + 1$. Then $x'' = u'v + uv'$.

[8] This is the cosine double-angle identity; it rewrites the combination $2\cos^2 t - 2\sin^2 t$ as $2\cos(2t)$.